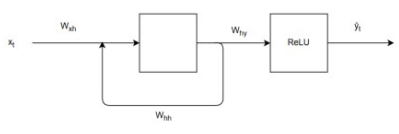
Brac University
CSE427: Machine Learning
Summer 2026
Quiz 05
Set A

Duration: 35 minutes

Name:	ID:	Section:

Question 1: **[4]**



Loss (SSR) = $\sum_{i=1}^{t} (y_i - \hat{y}_i)^2$

Assume, you have five data points of a sequence and you are trying to train the above RNN architecture.

Find the expression for $\frac{dLoss}{dW_{xh}}$ when the 5th datapoint (x_5) will be passed.

Question 2: **[4+2]**

A CNN is applied to a RGB image of size 32×32 and has the following layers, in order.
Layer 1: CONV 3x3, stride 1, padding 2, 16 filters
Layer 2: CONV 3x3, stride 1, padding 0, 32 filters
Layer 3: POOL 2x2, stride 2

a. **Find** the output size at each layer and total number of parameters in the whole network.
b. **What** will be the dimensions of the image matrix, the filter matrix, and the resulting output matrix of the convolution at Layer 2.

Question-1

$\frac{dLoss}{dW_{xh}} = \frac{dLoss}{d\hat{y}_5} \times \frac{d\hat{y}_5}{dW_{xh}}$ | Loss (SSR) $= \sum_{i=1}^{t} (y_i - \hat{y}_i)^2$

$\frac{dLoss}{d\hat{y}_5} = \frac{d}{d\hat{y}_5}\left[\sum_{i=1}^{t} (y_i - \hat{y}_i)^2\right]$

$= -2 \sum_{i=1}^{t} (y_i - \hat{y}_i)$ or $-2(y_5 - \hat{y}_5)$

[for all other y the differentiation = 0]

$\hat{y}_5 = ReLU(z_5)$

$z_5 = ((((x_1 W_{xh} W_{hh} + x_2 W_{xh}) W_{hh} + x_3 W_{xh}) W_{hh} + x_4 W_{xh}) W_{hh} + x_5 W_{xh}) W_{hy}$

$= (x_1 W_{xh} W_{hh}^4 + x_2 W_{xh} W_{hh}^3 + x_3 W_{xh} W_{hh}^2 + x_4 W_{xh} W_{hh} + x_5 W_{xh}) W_{hy}$

Now, $\frac{d\hat{y}_5}{dW_{xh}} = \frac{d\hat{y}_5}{dz_5} \times \frac{dz_5}{dW_{xh}}$

$\frac{d\hat{y}_5}{dz_5} = \frac{d}{dz_5} ReLU(z_5) = ReLU'(z_5)$

$\frac{dz_5}{dW_{xh}} = \frac{d}{dW_{xh}} (x_1 W_{xh} W_{hh}^4 + x_2 W_{xh} W_{hh}^3 + x_3 W_{xh} W_{hh}^2 + x_4 W_{hh} W_{hh} + x_5 W_{xh}) W_{hy}$

$= (x_1 W_{hh}^4 + x_2 W_{hh}^3 + x_3 W_{hh}^2 + x_4 W_{hh} + x_5) W_{hy}$

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$\therefore \frac{dLoss}{dW_{xh}} = -2 \sum_{i=1}^{t} (y_i - \hat{y}_i) \times ReLU'(z_5) \times (x_1 W_{hh}^4 + x_2 W_{hh}^3 + x_3 W_{hh}^2 + x_4 W_{hh} + x_5) W_{hy}$

Question-2

[a] Given, RGB image of size 32X32

So, image dimension 32X32X3 [RGB → Channel=3, Grayscale → 1 = 1]

So, W = H = 32

Layer-1 conv 3×3, stride, S=1, padding, P=2

No. of filter = 16

Output = $\frac{W - F + 2P}{S} + 1 = \frac{32 - 3 + 2\times2}{1} + 1 = 34$

So, Feature map dimension 34 X34 X16 (as W=H=32) → No. of filter

Parameters = Dimension of filter × No. of filter + No. of filter

$= (3 \times 3 \times 3) \times 16 + 16$

$= 448$ [conv 3×3, image channel=3, so, filter channel=3]

Layer-2 conv -3×3, stride 1, padding 0, 32 filters.

Output size = $\frac{34 - 3 + 2\times0}{1} + 1 = 32$ [Last layer output 34×34×16, W=H=34]

So, Feature Map dimension 32×32×32 → No. of filter

Parameters = $(3 \times 3 \times 16) \times 32 + 32 = 4640$

[conv -3×3] [image/output was 34×34×16, so, Filter channel = 16]

Layer-3 POOL 2×2, stride, S=2 [there is no padding in Pooling Layer]

Output size = $\frac{W - F}{S} + 1$

$= \frac{32 - 2}{2} + 1 = 16$ [previous layer output 32X32X32, W=H=32]

So, Feature map dimension 16 ×16 × 32

Parameters = 0 [Pooling Layer has no Learnable weight]

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So, Total parameter = 448 + 4640 + 0 = 5088

[b] At Layer-2,

input dimension 34×34×16

Filter dimension 3×3×16

Output Feature map dimension 32×32×32

To No. of filter = 32

Image Matrix

output dimension 32×32×32 → channel

So, No. of blocks in image Matrix

= 32×32 [blocks in each channel]

= 1024

Filter dimension = 3×3×16

So, No. of pixels in each block = 3×3×16

= 144

So, Image Matrix size = 144 × 1024

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Filter Matrix

No. of pixels in Each Filter = 3×3×16

= 144

[Same as No. of pixels in each block]

No. of Filters in Layer-2 = 32

So, Filter Matrix Size = 32×144

Output Matrix

output Matrix = (Filter Matrix) × (Input Matrix)

= (32×144) × (144×1024)

= 32 × 1024

= 32 × 32 × 32, [Reshaped]

↓ dimension

No. of Channel

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